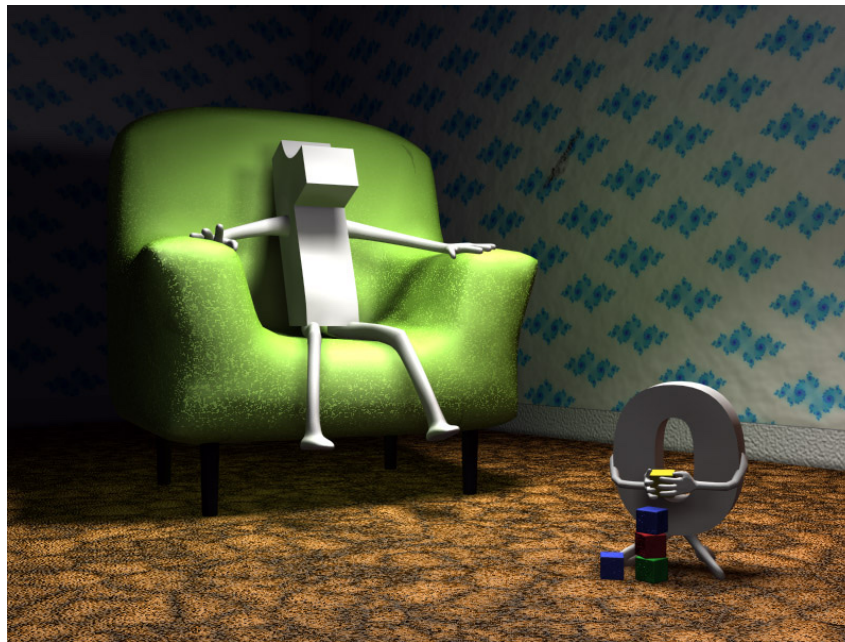


Chapter 1. INTEGERS

1. 2 Divisibility and Prime Factorisation

1.2.2 Gcd. Prime and coprime integers



Today we will prove a very important result:

BEZOUT'S LEMMA

BÉZOUT'S LEMMA. Let m and n be two integers, not both zero. Then $d = \gcd(m, n)$ exists and $d = xm + yn$ for some integers x and y .

Proof. Consider

$$\underline{X} = \{ sm + tn \mid s, t \in \mathbb{Z}, sm + tn \geq 1 \}.$$

Notice that $\underline{X} \neq \emptyset$ because $m^2 + n^2 \in \underline{X}$.

By the Well-Ordering Axiom, $\underline{X}$ has a least integer, say d . Then $\underline{d} \geq 1$ and $d = xm + yn$ for some integers x and y .

Claim $d = \gcd(m, n)$.

If $k \mid m$ and $k \mid n$, then $k \mid d$. By Theorem 1.10 (4), we have that

$$k \mid (xm + yn), \text{ that is, } k \mid d$$

It remains to show that $d \mid m$ and $d \mid n$.
Let us prove that $d \mid n$. Applying the
Division Algorithm, we have that

$$n = qd + r, \text{ where } 0 \leq r < d.$$

We have that

$$\begin{aligned} \underline{r} &= n - qd = \underset{\substack{\uparrow \\ d = xm + yn}}{n - q(xm + yn)} = \\ &= \underline{(-qx)m + (1 - qy)n} \end{aligned}$$

If $r > 0$, then $r \in \mathbb{X}$, but this
is impossible because $r < d$, and d is the
least element of $\mathbb{X}$. So $r = 0$,
and so $n = qd$, which shows
that $d \mid n$. ■

COROLLARY 1.15. Two integers m and n are relatively prime if and only if there exist integers x and y such that $xm + yn = 1$. In particular, any two consecutive integers k and $k + 1$ are relatively prime.

Proof.

$\Rightarrow$ | Bézout's Lemma

$\Leftarrow$ | Suppose that $xm + yn = 1$ and let us show that $\gcd(m, n) = 1$. To do so, we will prove that the only positive common divisor of m and n is 1.

If $d \geq 1$ and $d \mid m$ and $d \mid n$.

Then by Theorem 1.10 (4), $d \mid (xm + yn)$,

so $d \mid 1$. Thus: $d = 1$



COROLLARY 1.16. If $d = \gcd(m, n)$, then m/d and n/d are relatively prime.

Proof By Bézout's Lemma we have that
 $d = xm + yn$ for some integers x and y .

But then $1 = x\left(\frac{m}{d}\right) + y\left(\frac{n}{d}\right)$.
The result now follows from the
previous corollary. $\square$

Remark. $d > 1$, $d = xm + yn$ for some integers x, y $\xLeftrightarrow{\text{Bézout's Lemma}}$ $d = \gcd(m, n)$
 $\nRightarrow$

The converse of Bézout's Lemma does not hold for $d > 1$.

$$m = 4 \quad \text{and} \quad n = 8$$

$$12 = 1 \cdot 4 + 1 \cdot 8$$

↓

But $12 \neq \gcd(4, 8)$. Because

$$\gcd(4, 8) = 4$$
